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# REFLECT Project — Student Guide

## 1. What is this project?

You will design a small interactive process that helps a student better understand:

- what happened
- why it happened
- what to do next

The focus is on helping someone make sense of a messy or confusing experience.

## Current situation and context

From my perspective as someone working with students, I often notice that students:

- remember situations only partially
- describe events without really understanding them
- jump quickly to conclusions
- struggle to turn insight into action

As a result, reflection can become writing about experience without actually improving understanding.

36 The problem is that thinking is often *invisible, unstructured, and difficult to revisit over time*.

37 This project explores how interaction design could help support that process.

## 38 Core idea

39 You are designing a tool that helps someone:

- 40 • reconstruct experience
- 41 • organize thoughts
- 42 • explore explanations
- 43 • notice patterns
- 44 • make decisions

45 The goal is helping people understand their own experience more clearly.

46

## 47 Design directions

48 Below are different ways of approaching the problem. You do **not** need to combine  
49 everything. Choose one direction that interests you and explore it deeply.

### 50 A. Memory & Capture

51 Sometimes the problem starts before reflection even happens.

52 People forget details, lose context, or only remember fragments.

53 This direction explores:

- 54 • capturing moments while they are fresh (audio?, text, trackers,...)
- 55 • organizing messy thoughts (tags, color-schemes, mind-maps,...)
- 56 • helping users reconstruct situations later (Notebook LM styl retrieval and question  
57 answering – Retrieval augmented generation)

58 Possible questions:

- 59 • How can someone quickly capture a moment?
- 60 • What information becomes important later?
- 61 • How can fragmented memories become usable?
- 62 • Is the user talking about a structural point (trait) or just transient events?
- 63 • What is the magnitude, severity, or size of the issue? What is the saliency of the point  
64 being discussed?

65

### 66 B. Structure & Visualization

67 Many reflections stay as long blocks of text.

68 But text alone can make it difficult to see relationships, sequences, or causes.

69 This direction explores:

- 70 • timelines
- 71 • visual maps
- 72 • spatial organization
- 73 • connections between events

74 Possible questions:

- 75 • How can structure become visible?
- 76 • How can users see what led to what?
- 77 • How can experiences become easier to overview?

78

## 79 C. Reasoning & Interpretation

80 Students often stop at their first explanation.

81 This direction explores tools that support deeper thinking and *exploration*.

82 Examples:

- 83 • comparing explanations
- 84 • checking assumptions
- 85 • exploring alternatives
- 86 • organizing evidence

87 Possible questions:

- 88 • How can a system encourage better reasoning?
- 89 • How can users test their own interpretation?
- 90 • How can uncertainty become visible?
- 91 • Would pre-defined questions work, or do we need adaptation (AI asking questions)?
- 92 • Which system is best? Socratic questioning, Gibbs, ...

93

94

## 95 D. Entities & Relationships

96 Experiences often involve recurring things:

- 97 • people
- 98 • projects
- 99 • meetings
- 100 • feedback
- 101 • tools
- 102 • themes

103 This direction explores ways to make these visible across time. Possible questions:

- 104 • What keeps appearing in someone's experience?
- 105 • How are situations connected?
- 106 • Can users explore relationships between people, events, or themes?
- 107 • How should we handle AI transcription errors (what interface is needed for entity
- 108 alignment)

109

## 110 AI as support

111 In this project, AI is not the “thinker.”

112 Instead, AI can act as a support layer.

113 For example:

- 114 • cleaning messy notes
- 115 • helping organize information
- 116 • asking clarification questions
- 117 • retrieving earlier notes or memories

118 The important design challenge is:

119 How do we support human thinking without replacing it?

## 120 Example interaction ideas

121 These are examples for inspiration only.

122 You do not need to copy them.

123 Examples:

- 124 • a timeline reconstruction tool
- 125 • a visual relationship map
- 126 • a guided reflection conversation

- 127 • an AI-assisted note cleanup view
- 128 • a hypothesis-testing interaction
- 129 • a memory capture flow
- 130 • a drag-and-drop reasoning canvas

## 131 What good work looks like

132 Strong projects usually:

- 133 • focus on one clear interaction
- 134 • help users understand something new
- 135 • make thinking more visible
- 136 • reduce confusion or overload
- 137 • lead toward meaningful action
- 138 • are tested with real users or realistic situation

139

## 140 Theory background

141 This project draws on three complementary perspectives from cognitive science and learning  
142 theory that together help explain why reflection is difficult and how interaction design can  
143 support it. These theories are not used to build a psychological model of reflection, but as  
144 design lenses: they help us think about how interfaces, structure, and interaction can either  
145 support or interrupt reflective thinking.

146 In simple terms, the following ideas were taken from these theories:

- 147 • **Dewey:** avoid fixed interpretation → the tool should support evolving inquiry
- 148 • **Dual-process theory:** avoid immediate conclusions → the tool should support  
149 reconsideration and conflict detection
- 150 • **Cognitive Load Theory (CLT):** avoid unnecessary mental effort → the tool should  
151 support externalized structure and memory

## 152 Reflection as inquiry

### 153 Dewey, J. (1910). *How We Think*

154 In Dewey's view, reflection is not passive thinking but an active process of inquiry grounded  
155 in experience. Reflection begins when a person encounters a situation that feels uncertain,  
156 incomplete, or confusing. From there, the person starts questioning, interpreting, and  
157 revisiting the experience in order to better understand it.

158 Importantly, reflection is tied to action. The goal is not simply describing what happened, but  
159 reorganising experience into something that can guide future decisions and behaviour.  
160 Reflection therefore becomes a process of making sense of experience rather than merely  
161 recording it.

162 This perspective is especially relevant to the project because students often describe  
163 situations without fully examining why they happened or how they connect to broader  
164 patterns over time. The project therefore explores how interaction design can support inquiry  
165 rather than just documentation.

### 166 A more modern interpretation:

167 Rodgers, C. (2002). *Defining reflection: Another look at John Dewey and reflective thinking*.  
168 Teachers College Record, 104(4), 842–866.

169 <https://journals.sagepub.com/doi/10.1111/1467-9620.00181>

170

## 171 Dual-process theory and cognitive friction

172 Dual-process theories of cognition distinguish between fast, intuitive thinking (System 1) and  
173 slower, more deliberate reasoning (System 2). Everyday interpretation of experiences often  
174 relies on System 1, producing quick but sometimes oversimplified explanations. System 2 is  
175 more likely to become engaged when something feels uncertain, conflicting, inconsistent, or  
176 effortful.

In this sense, cognitive friction — such as ambiguity, contradiction, or alternative explanations — can trigger deeper reflection by encouraging users to reconsider their initial interpretation rather than accepting it immediately.

For this project, this is important because students may quickly settle on simple explanations without exploring assumptions, evidence, or alternative perspectives. Interaction techniques such as comparison, questioning, visual organisation, or evidence mapping may help support more deliberate reasoning.

At the same time, the theory also suggests that too much friction can become counterproductive if the interaction becomes exhausting or overly complicated.

## Sources

- **Primary source (historical / foundational):**  
Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.  
→ System 1 = fast intuition, System 2 = effortful reasoning
- **Foundational academic paper:**  
Stanovich, K. E., & West, R. F. (2000).  
*Individual differences in reasoning: Implications for the rationality debate?*  
Behavioral and Brain Sciences, 23(5), 645–665.  
→ Introduces dual-process distinctions in cognitive psychology
- **Modern interpretation (conflict triggering deeper reasoning):**  
Evans, J. St. B. T., & Stanovich, K. E. (2013).  
*Dual-process theories of higher cognition: Advancing the debate*. Perspectives on Psychological Science, 8(3), 223–241.  
→ System 2 is more likely to engage when conflict, uncertainty, or monitoring demands arise

## Cognitive Load Theory and external support

Cognitive Load Theory emphasises that working memory is limited, and that learning and reasoning suffer when too much mental effort is spent on irrelevant or poorly structured information. The theory distinguishes between:

- **Intrinsic load** → the complexity of the task itself
- **Extraneous load** → unnecessary effort caused by poor design or interaction
- **Germane load** → productive mental effort that supports understanding and learning

For reflection tools, this suggests a need to reduce unnecessary mental effort by offloading memory, externalising structure, and helping users organise information more clearly.

For example, reflection can become difficult if users must simultaneously remember events, organise notes, interpret meaning, and navigate a complex interface. Timelines, visual maps, retrieval systems, and structured overviews may help reduce this burden by making information easier to revisit and interpret.

214 Importantly, CLT does not suggest removing all difficulty. Reflection still requires effort.  
215 The challenge is therefore to reduce unnecessary cognitive burden while preserving the  
216 productive effort involved in interpretation and meaning-making.

217 Here are some questions that can help with increasing the Germane load:

- 218 • Ask WHY not only WHAT:
  - 219 ○ Why do you think this approach worked in this situation?
- 220 • Go from concrete to abstract:
  - 221 ○ What general rule can you extract from this experience?
- 222 • Compare and contrast:
  - 223 ○ How was this situation different from the last time you faced something
  - 224 similar?
- 225 • Error Analysis:
  - 226 ○ What assumption turned out to be wrong?
- 227 • Make a prediction, and compare with the later outcome:
  - 228 ○ Step 1 (Before): What do you expect will happen? Why?
  - 229 ○ Step 2 (after): Where did your prediction differ from reality?

## 230 Sources

- 231 • Sweller, J. (1988).  
232 *Cognitive load during problem solving: Effects on learning*. Cognitive Science, 12(2),  
233 257–285.
- 234 • Sweller, J., van Merriënboer, J. J. G., & Paas, F. (2019).  
235 *Cognitive architecture and instructional design: 20 years later*. Educational  
236 Psychology Review, 31, 261–292.

## 237 Implications for design

238 Together, these perspectives suggest that reflective systems must balance three goals:

- 239 • supporting open-ended inquiry and meaning-making (Dewey)
- 240 • encouraging deeper reasoning when appropriate (dual-process theory)
- 241 • reducing unnecessary cognitive burden (CLT)

242 The design challenge is therefore not simply to “help users reflect,” but to balance ease of use  
243 with moments of productive difficulty that support interpretation and learning.

## 244 Critical design questions

245 What would each theory say my design is doing wrong?

- 246 • **Dewey** → am I killing inquiry by structuring too early?
- 247 • **Dual-process theory** → am I failing to trigger deeper reasoning?
- 248 • **CLT** → am I overloading working memory with unnecessary UI complexity or
- 249 interaction steps?